Belief-State Control for Diplomatic Negotiation with a Lower-Cost Language Model

Anonymous ACL submission

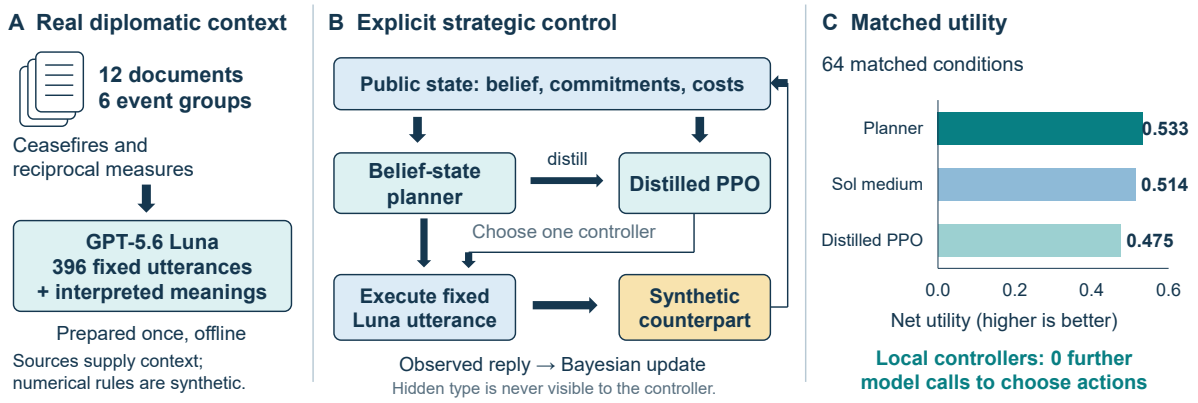


Figure 1: Real diplomatic sources ground fixed Luna utterances. A public-belief planner or its distilled PPO policy selects actions under the same language execution. Responses update public beliefs; distillation transfers planner supervision. Zero further model calls refers only to local strategy selection.

Abstract

Can explicit strategic control improve negotiation with a lower-cost language model? We study a simulation grounded in 12 real diplomatic documents, holding GPT-5.6 Luna's generated utterances and their interpretations fixed across controllers. On 64 matched conditions, belief-state planning achieves net utility 0.533, compared with 0.514 for direct action selection by GPT-5.6 Sol with medium reasoning. Planner-distilled PPO reaches 0.475 without further model calls for strategy selection. Separate interventions attribute most of the learned policy's gain to signals and updated beliefs. Dialogue analysis connects these mechanisms to conditional offers and commitment wording. The results identify the benefits of explicit control under declared negotiation rules.

1 Introduction

Large language model (LLM) agents create opportunities to negotiate on behalf of users (Yao et al., 2023; Bianchi et al., 2024). They must pursue user goals while responding to parties with different interests and incomplete information. Existing benchmarks show that strategic choices affect outcomes (Abdelnabi et al., 2024). Repeated decisions also make inference cost relevant (Kapoor et al., 2024), motivating systems that assign specialized roles to lower-cost models (Belcak et al., 2025).

Prior work separates strategy from language generation (He et al., 2018), combines dialogue with planning (Meta Fundamental AI Research Diplomacy Team (FAIR), 2022), and improves smaller negotiators through reinforcement learning and distillation (Hua et al., 2026; Tabacof and Joglekar, 2026). Building on these approaches, we ask: with language realization held fixed, how does explicit strategic control affect negotiation utility and the cost of choosing actions?

We construct a diplomatic negotiation simulation from 12 real documents across six event groups. The documents provide context and examples of reciprocal conditions; utilities, hidden opponent types, and responses are synthetic. GPT-5.6 Luna (Luna) generates and separately interprets 396 fixed utterances. We compare Random, Heuristic, Bandit, PPO, direct model decisions, an exact public-belief planner, and a policy distilled from that planner (Figure 1). The planner knows the simulation rules but never observes the hidden type.

On 64 matched conditions, the planner outper-

forms direct action selection by GPT-5.6 Sol (Sol) with medium reasoning. Distilled PPO improves on base PPO without further strategy-selection model calls. Interventions and dialogue traces distinguish information-seeking questions from incentive-changing signals and test whether actions preserve expressed commitments. Our contribution links negotiation outcomes, strategic computation, and discourse meaning under controlled language conditions.

2 Related Work

Negotiation agents. Lewis et al. (2017) learn from human bargaining dialogues and improve decisions through simulated continuations. Recent benchmarks cover resource allocation, trading, and multiple stakeholders (Bianchi et al., 2024; Abdelnabi et al., 2024). CICERO combines dialogue and strategic reasoning in the board game Diplomacy (Meta Fundamental AI Research Diplomacy Team (FAIR), 2022). Our simulation uses real diplomatic documents to frame reciprocal measures and commitments.

Strategic control. He et al. (2018) separate dialogue acts from linguistic realization. GDP-Zero uses LLM simulations for tree search (Yu et al., 2023); PPDPP learns a separate policy planner (Deng et al., 2024); and Dual-Mind combines strategy planning with expression refinement (Liu et al., 2025). We build on this separation by holding utterances and interpretations fixed when comparing controllers, then analyzing their information use and commitments.

Efficient agents. Kapoor et al. (2024) advocate evaluating performance alongside cost, while Belcak et al. (2025) argue for specialized small-model roles. SocialRL combines reinforcement learning and teacher distillation for negotiation (Hua et al., 2026). Tabacof and Joglekar (2026) study reinforcement learning across seller model sizes. We keep language model weights fixed and train a separate controller, measuring strategy-selection calls and examining the mechanisms behind utility gains.

3 Method

Data and task. We use 12 document contexts from six diplomatic event groups, including CMFA PressCon (Mochtak and Turcsanyi, 2021) and Security Council discourse. Training covers the Korean Peninsula, South Sudan, and Kashmir (six contexts); development uses Gaza (two); testing uses Syria and Iran (four). Passages about ceasefires, reciprocal measures, and commitments guide synthetic bilateral negotiations. One party offers to maintain its measure for $x \in \{1, 2, 3\}$ weeks if the counterpart maintains its measure for $y \in \{1, 2\}$ weeks. Durations, payoffs, types, and responses are researcher-defined.

State, actions, and reward. The agent has at most five decisions and cannot observe the counterpart's type, $\tau \in \{S, H\}$. It observes the offer, acceptance, commitments, costs, and belief $b_t = P(\tau = H)$. Eleven actions cover ambiguity, signaling, a private question, six conditional offers, ratification, and withdrawal. Questions provide noisy type cues; offer responses also reveal information. Ratification requires observed acceptance and own utility at least as high as the outside option.

Specifically, an agreement gives $u_A = 0.5y - 0.18x$, and the counterpart accepts when $u_B = 0.35x - c_\tau y + \delta z \geq 0$, where z records a signal and c_τ determines the cost of reciprocal measures. A signal therefore changes an incentive; a question changes information. For response likelihoods L_H and L_S, the public belief becomes

$$b_{t+1} = \frac{b_t L_H}{b_t L_H + (1 - b_t) L_S}. \quad (1)$$

Equal response likelihoods leave the belief unchanged. Step rewards are

$$r_t = d_t u_{\text{end}} - 0.01 - c_t^{\text{act}} - 0.005 e_{t+1}^2, \quad (2)$$

where d_t marks termination, e counts rejected offers, and c_t^{act} sums question, signal, rejection, reversal, and invalid-action charges. Terminal utility is u_A at agreement or 0.12 at withdrawal or expiration, paid once. We report $\sum_t r_t$. Appendix A gives all costs and transition rules.

Shared language execution. GPT-5.6 Luna generates three styles per context–action pair, yielding 396 utterances. A separate Luna call interprets their function, reciprocal terms, and stated cost. These meanings determine transitions; interpretation mismatches are retained. All controllers share these utterances and templated counterpart replies, separating action selection from language realization.

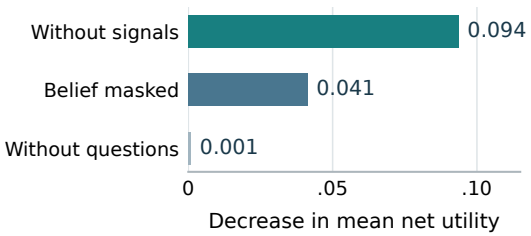


Figure 2: Evaluation-time interventions on distilled PPO. Losses are differences in three-seed mean utility on the follow-up conditions. Full values and seed variation appear in Table 3.

Controller	Utility	Agree.	Calls
A. Initial: 256 conditions, three styles			
Random	-0.146 ± 0.001	11.0	0
Bandit	0.118 ± 0.001	23.8	0
Heuristic	0.419	94.8	0
PPO	0.358 ± 0.018	73.0	0
Planner	0.488	93.3	0
B. Matched: 64 conditions, one style			
Luna none	-0.368	24.3	3.949
Sol none	-0.367	26.3	4.773
Sol medium	0.514	100.0	3.429
Heuristic	0.470	100.0	0
PPO	0.378 ± 0.012	81.9	0
Distilled PPO	0.475 ± 0.016	98.3	0
Planner	0.533	100.0	0

Table 1: Policy comparison with fixed Luna utterances. Agreement is a percentage; calls are per negotiation for strategy selection. ±: three-seed SD (evaluation seeds for Random). Panels use different conditions.

Strategic controllers. We compare Random, a development-selected signal-first Heuristic, an immediate-reward Bandit, and PPO (Schulman et al., 2017). An exact belief-state planner maximizes expected net utility under the declared rules without seeing the true type. We distill its action values from training contexts and configurations, then refine the policy with PPO. Direct Luna (reasoning `none`) and Sol (`none`/`medium`) receive the public state and numerical rules.

The planner solves response branches by backward recursion. Distillation uses 43,440 public states from planner and exploratory trajectories, learning a soft distribution over action values (temperature 0.03) and a value estimate. After 2,400 supervised updates, PPO refines the network through interaction. At deployment, it selects the highest-scoring action from 31 public features, replacing repeated planning with local inference.

4 Experimental Setup

The initial evaluation crosses four test contexts with 64 numerical configurations and three styles. A matched comparison uses the first 16 configurations and one style, giving 64 context–configuration conditions. A follow-up uses 64 new configurations with the same test contexts and three styles. We compute expectations over hidden types and question responses, then average within event groups and across groups. Learned results average three seeds. Bandit and base PPO each receive 102,400 training steps per seed; distilled PPO receives additional teacher supervision. Final checkpoints are used, with method selection on development data. Appendices A and C provide settings and diagnostic analyses.

5 Results

Initially, PPO outperforms Random and Bandit but falls below the heuristic (Table 1). In the matched comparison, the planner reaches 0.533 utility versus 0.514 for Sol medium. Distilled PPO reaches 0.475 versus 0.378 for base PPO and 0.470 for the heuristic. Sol none remains near Luna none: direct-model performance depends strongly on reasoning settings.

Local controllers require no language model calls for action selection. Sol medium uses an expected 3.429 calls per negotiation, corresponding to 4,184.8 tokens and $0.02363 at the recorded request costs. These figures count logical requests even when experimental caching avoids a repeated call. These strategy-selection figures exclude shared utterance construction, interpretation, and offline training.

On the follow-up configurations, distilled PPO achieves 0.468 utility, compared with 0.364 for base PPO and 0.445 for the heuristic (Figure 2). Blocking signals reduces utility by 0.094; replacing the posterior feature with the prior reduces it by 0.041. Blocking questions reduces it by only 0.001. These are evaluation-time interventions without retraining. The learned policy benefits mainly from signals and updated beliefs in this setting. Updated beliefs can come from offer responses as well as questions. The planner's no-question comparison yields a larger utility loss of 0.008.

Discourse analysis. Matched Iran-context traces illustrate this distinction (Figure 3). Luna none re-



Figure 3: Selected paths under the same Iran-context condition and soft counterpart. Terms $x{:}y$ denote own and reciprocal durations. Values are episode net utility; the learned path uses seed 17.

peats a rejected 1:2 offer four times and withdraws, yielding -0.80. The planner and Sol medium ask a private question, send a signal, and obtain acceptance of that same offer, yielding 0.70. The question changes the belief from 0.65 to 0; the signal raises the soft counterpart's acceptance utility from -0.05 to 0.05. The learned controller instead signals and offers 2:2, yielding 0.55. Wording also matters: in a Syria-context pair, explicit withdrawal terminates the episode, while conditional withdrawal is interpreted as invalid and incurs repeated costs. Appendix B examines these paths and distinguishes expressed readiness to ratify from ratification itself.

Both types reject the unsignaled 1:2 demand, so repetition provides no new information. A question resolves uncertainty; the subsequent signal makes the demand acceptable. The learned policy instead concedes more. Successful agreement therefore does not establish that it reproduces the planner's information-gathering strategy.

6 Discussion

Explicit control improves decisions when negotiation rules can be specified. The planner accounts for beliefs, costs, and future responses. Distillation transfers part of this behavior to a policy without online search or model calls. Its remaining gap to Sol medium and small advantage over the heuristic distinguish exact planning from learned control.

Signals drive more of the learned gain than questions. Offer responses can also update beliefs, so information value depends on subsequent choices and acquisition costs. Discourse analysis adds a separate requirement: utterances must preserve the selected conditions and commitment strength. Evaluation should connect these meanings to utility and information use.

7 Conclusion

With Luna utterances held fixed, belief-state planning achieved the highest matched utility, and planner-distilled PPO improved on base PPO without further model calls for action selection. Signals, updated beliefs, and commitment wording explain different parts of the observed behavior. This controlled study identifies explicit strategic control as a useful component for source-grounded diplomatic negotiation simulations.

Limitations

Real documents ground the task, but payoffs and dialogues are synthetic. Evaluation covers four contexts from two event groups; extra numerical settings are not independent historical cases. Fixed utterances limit claims about online dialogue. All 44 matched style-0 meanings equal canonical templates, so utility does not establish a generation advantage. Shared-model interpretation lacks independent human validation, and some commitment expressions are misclassified.

The planner knows the response rules and interpretation mapping; distillation adds teacher supervision and computation. These resources limit comparisons with model-free learning. Model mismatch remains untested, and the tested reasoning settings do not isolate model-size effects. Three seeds and one cached model response per state do not capture historical or model-output uncertainty. Cost comparisons cover strategy-selection calls, excluding language preparation, teacher and training costs, and local computation. Detailed assumptions and procedures appear in Appendices A–B.

Ethical Considerations

Public statements provide context, not private preferences or predictions of state behavior. Synthetic payoffs should not guide real diplomatic decisions. Source provenance is preserved; supplied-corpus access remains restricted to the collaboration. Language models assisted utterance generation, interpretation, drafting, and a visual concept. The final teaser was redrawn from recorded results. Model checks are not independent human annotation.

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A Implementation and Reproducibility

Environment. The soft-type cost is $c_S = 0.2$. A first question yields a type cue with accuracy q; later questions give no new cue. Each turn costs 0.01, each question c_q, and each rejection c_r, plus the escalation cost in the main text. Signals incur their interpreted expenditure (intended 0.06), set $z = 1$, and do not stack. Reducing an offered duration below the recorded commitment costs 0.08. Ambiguity makes the next offer provisional without erasing prior commitments. Invalid interpretations or ratifications cost 0.10. Ratification requires observed acceptance and $u_A \geq 0.12$; withdrawal or expiration pays 0.12 before costs.

We draw 96 training, 32 development, 64 initial test, and 64 new follow-up configurations over priors, question reliability/cost, rejection cost, signal effect, and hard-type cost. Exact grids are released. The 31 public features exclude source identity and hidden type. All 17 mismatches among 396 intended/interpreted action pairs remain; these are automatic checks, not human labels.

Training and aggregation. PPO uses two 64-unit layers, learning rate 3×10^{-4}, discount .95, entropy .01, rollout 256, batch 64, and ten epochs. Bandit exploration is .2. Both use seeds 17, 29, and 43. Distillation uses Adam at .001, batches of 256, value-loss weight .5, and gradient clipping .5; 35% random exploration augments teacher trajectories. Teacher computation is additional to PPO interaction. Evaluation averages styles and configurations within event groups, then weights groups equally. SD is across seeds, not historical events. Follow-up interventions appear in Table 3. Single-seed training ablations yield .346/.327 for discount .8/.99, .366/.392 for entropy 0/.1, and .377/.371 without rejection/reversal costs.

Released detail. Source sentences and provenance are in `study_results/contexts.json`. The repository includes traces, checkpoints, reproduction commands, and the unabridged supplement in `paper/supplementary_appendix_full.tex`.

B Six Discourse Cases

These are synthetic, source-grounded episodes, not historical dialogues. Cases 1–4 share Iran context `ir1` (UNSC S/2020/644, 30 June 2020), condition 0, style 0, and the soft-type evaluation branch. Cases 5–6 share Syria context `sy1` (CMFA briefing, 1 March 2016), condition 25, and the hard-type branch. Hidden types are not controller inputs.

1. Luna none. "We will suspend the specified sanctions for 1 week, conditional upon your suspension of the specified nuclear activities for 2 weeks" is rejected and repeated three times before withdrawal. Both types reject this unsignaled offer, so belief stays .65. Repetition adds costs without information; net utility is $-.80$.

2. Planner. "Privately, we ask whether you require a larger reciprocal concession" elicits a soft response, reducing belief to zero. A signal then changes counterpart utility for the 1:2 offer from $-.05$ to .05. Question, signal, offer, and ratification yield .70: the question supplies information; the signal changes incentives.

3. Sol medium. Sol follows the same path and shared utterances, also yielding .70. Its final utterance starts "We reaffirm our readiness to ratify the currently accepted agreement". The interpreter records ratification, although readiness does not itself perform that act. This is not independent language generation.

4. Distilled PPO, seed 17. The policy signals and offers "We will suspend the specified sanctions for two weeks, provided that you suspend the specified nuclear activities for the same period". Acceptance reveals the soft type; ratification follows. Utility is .55. It avoids the question but makes a larger concession. The same readiness-to-ratify mismatch applies.

5. Explicit withdrawal. After a signal and rejected 2:2 offer, the heuristic says "We hereby withdraw from these negotiations, without proposing or ratifying any agreement". The episode ends at $-.14$. Rejection reveals infeasibility; the signal is wasted because its configured effect is zero.

6. Conditional withdrawal. With another style, the same policy instead says "If these negotiations cannot advance without compromising our principles, we will withdraw without proposing or ratifying any agreement." The interpreter treats this conditional threat as invalid. Three attempts lead to expiration and $-.47$: three invalid charges and two additional turns and escalation charges explain the .33 loss.

C Diagnostic Results

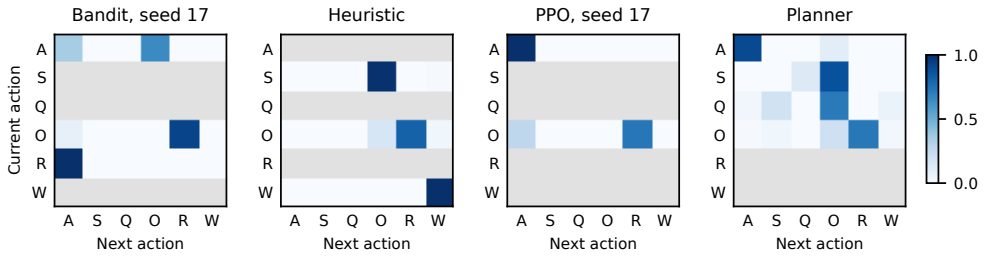


Figure 4: Row-normalized initial-test transitions. A: ambiguity; S: signal; Q: question; O: offer; R: ratify; W: withdraw. Grey: no outgoing counts. Learned panels: seed 17. Counts enumerate response paths.

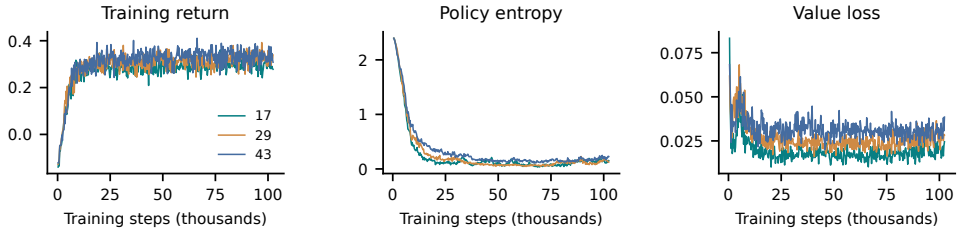


Figure 5: Base PPO training: rolling return, entropy, and value loss. Lower loss does not establish better negotiation: base PPO remains below the heuristic on the initial test.

Policy	Own	Other	Cost	Break	Turns	Max e	Mean e	Reverse	Viol.
Random	0.147	0.040	0.292	0.890	4.288	0.856	0.547	0.437	0.000
Bandit	0.140	0.082	0.022	0.762	1.773	0.030	0.017	0.000	0.000
Heuristic	0.531	0.260	0.112	0.052	3.148	0.160	0.119	0.000	0.000
PPO	0.413	0.278	0.055	0.270	2.459	0.264	0.263	0.000	0.000
Planner	0.570	0.176	0.082	0.067	3.030	0.211	0.183	0.000	0.000

Table 2: Initial-test metrics, event-group weighted. Own/Other: pre-cost utilities over all outcomes; Break: breakdown rate; e: rejection index; Reverse: reversals; Viol.: bilateral acceptance violations.

Controller / intervention	Utility	Change
PPO	$.364 \pm .016$	–
Heuristic	.445	–
Distilled only	$.446 \pm .011$	–
Distilled PPO	$.468 \pm .012$	–
Without questions	$.467 \pm .014$	$-.001$
Belief masked	$.427 \pm .013$	$-.041$
Without signals	$.374 \pm .014$	$-.094$
Planner	.500	–
Planner without questions	.492	$-.008$

Table 3: Follow-up: 256 conditions, three styles. $\pm$: three-seed SD. Changes use distilled PPO or full planner as reference. Policies are frozen; masking replaces posterior with prior.

PPO repeats ambiguity 2,260 times without gaining a type cue. Full counts and probabilities are in `transition_matrices.json`.

Changed rule	Planner	Questions	PPO
$q = 0.50$	0.527	0.00	0.410
$q = 0.70$	0.528	0.00	0.410
$q = 0.90$	0.532	1.00	0.410
$q = 1.00$	0.555	1.00	0.410
$\delta = 0.00$	0.486	1.00	0.410
$\delta = 0.10$	0.532	1.00	0.410
$\delta = 0.25$	0.596	1.00	0.410

Table 4: One-factor rule shifts: utility and expected planner question count. PPO averages three frozen seeds. Other rules stay fixed.

Base PPO uses neither questions nor signals here. Reliable information benefits the planner. Full sweeps are in `sensitivity_metrics.json`.